biosphere—anywhere that supports life, from the ocean depths to the highest mountaintops.

The biosphere's origins are ancient: fossils of tiny single-celled microorganisms that date back 4.28 billion years suggest that it is almost as old as Earth itself. The biosphere extends into every land- and water-based environment, and reaches into extreme habitats, such as the intensely hot mineral-rich waters around hydrothermal vents. It is often divided into "biomes"—common major habitats, such as deserts, grasslands, oceans, tundra, and tropical rain forests.

"Man is becoming a more and more powerful geological force, and the change of his position on the planet coincided with this process."

Vladimir Vernadsky

Earth the superorganism

Ideas about the biosphere began to emerge in the 18th century, when the Scottish geologist James Hutton described Earth as a superorganism—a single living entity. A century later, Eduard Suess introduced the concept of the biosphere in *Das Antlitz der Erde* (*The Face of the Earth*). Suess explained that life is limited to a zone at Earth's surface and that plants are a good example of the interactions between the biosphere and other zones—they grow in the soil of the lithosphere, but their leaves breathe in the atmosphere.

In *The Biosphere* (1926), Russian geochemist Vladimir Vernadsky, who had met Suess in 1911, defined the concept in much more detail, outlining his view of life as a major geological force. Vernadsky was one of the first to recognize that atmospheric oxygen, nitrogen, and carbon dioxide result from biological processes, such as the respiration of plants and animals. He argued that living organisms reshape the planet as surely as physical forces, such as waves, wind, and rain. He also introduced the idea of three stages of